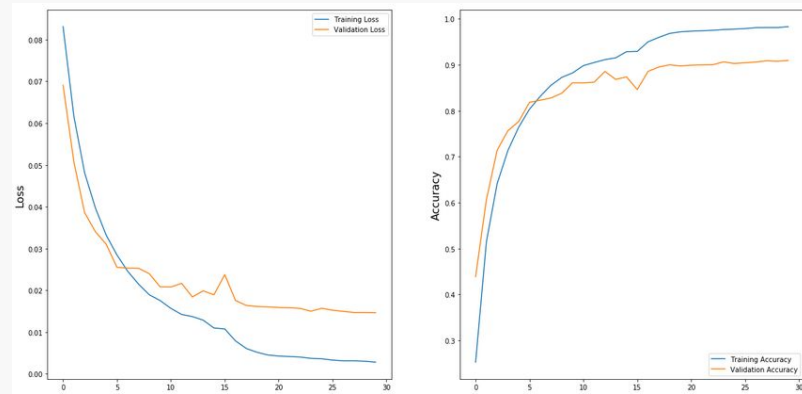


Modded Vision Transformer Distillation Benchmark

Ayush Tripathi, Sai Minnal, Mayank Deoras

Problem Statement & Motivation

- Deep learning projects often optimize for final accuracy
- Final accuracy by itself doesn't fully represent efficiency
- Compute cost & hardware efficiency matter
 - Particularly with diminishing returns
- We are working to minimize time required for a Vision Transformer (ViT) to reach 85% test accuracy on CIFAR-10



Standard accuracy and loss curves for model training against epochs — clear diminishing returns

Mathematical Formulation of the Problem

- ViT breaks images into N patches (P by P) $\rightarrow$ feature vector (vector of length d)

$$N = \left(\frac{H}{P}\right) \left(\frac{W}{P}\right)$$

- ViT adds positional embeddings (attention doesn't know order/location)

$$Z_0 = [z_{\text{cls}}; z_1; \dots; z_N] + E_{\text{pos}}$$

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- QK^T is an N by N matrix — attention cost grows like $O(N^2)$
 - Patch count N is a huge performance indicator

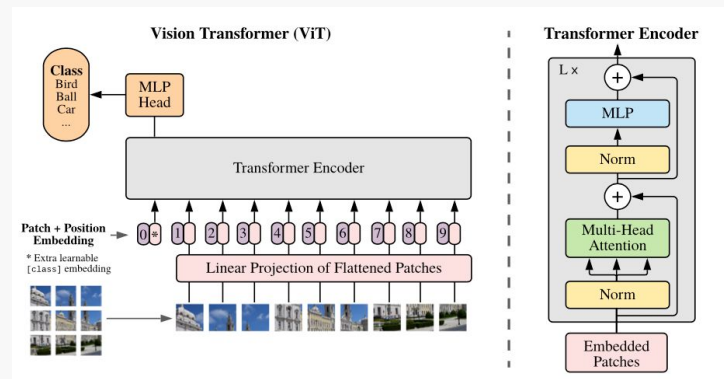


Illustration of the self-attention mechanism

First Approach: System Optimization

- System-level improvements

- AMP / bf16
- AdamW + cosine LR + warmup
- Data loading

- Measurements

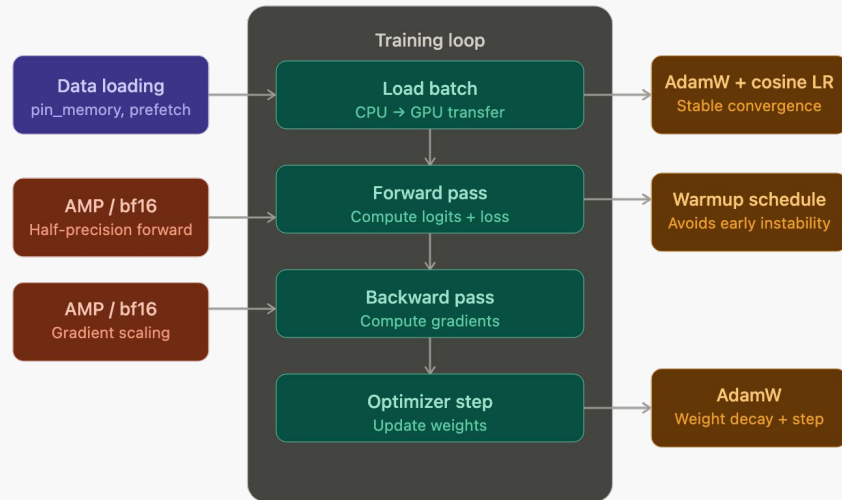
- Time-to-target (seconds to reach 85% accuracy)

- Observations

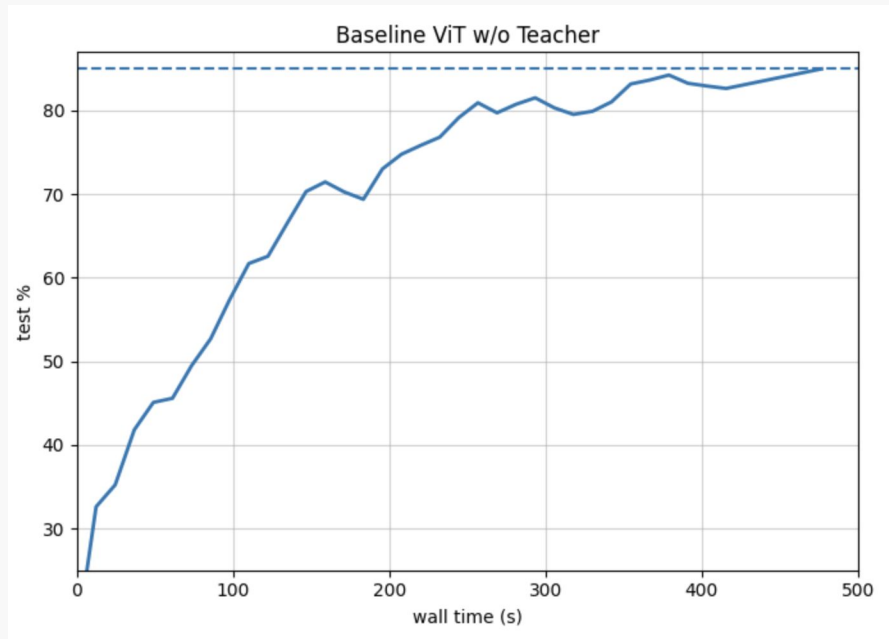
- Faster throughput and lower memory
- High time-to-target for 85%

- Conclusion

- Training becomes increasingly inefficient near higher accuracies
- System-level changes alone can't overcome this bottleneck



Initial Results of the First Approach



Baseline: reached 85.0% in 476.7s

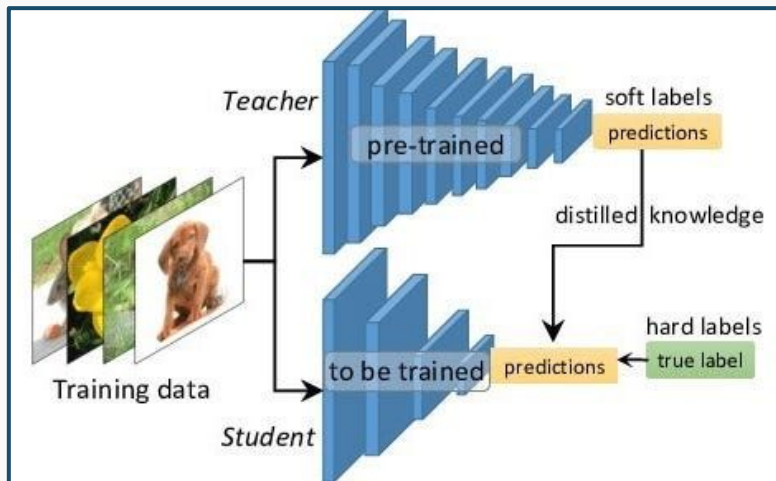
Pivoting to Distillation

Insight

System-level optimizations improved speed per step.

Convergence behavior is not optimized.

Using a ViT allows for more flexibility and power at scale compared to a CNN, but takes longer to train.



New Goal:

Minimize time required to reach 85% test accuracy on CIFAR-10 using various distillation methods.

Pivot

Bottleneck isn't just the system but how the model learns.

One-Hot Encoding leads to limitations.

CNN teacher-guided training provides richer, more informative supervision with faster convergence while retaining the benefits of the ViT.

Vanilla Knowledge Distillation (KD)

- **Setup**

- Teacher (ResNet-56, frozen) → Student (ViT)
- Train on labels and teacher predictions

- **Key Idea**

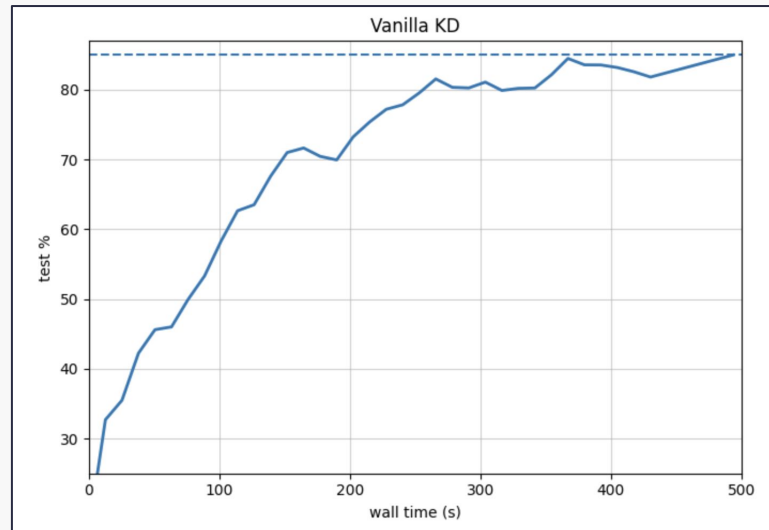
- Combine standard cross-entropy loss with hard labels and distillation loss between student and teacher logits
- Teacher provides soft targets (probability distributions)
- Captures class relationships (“dark knowledge”)

- **Benefits**

- Richer supervision than labels alone
- Faster convergence and smoother training

- **Limitation**

- Treats all non-target classes the same



KD reached 85.0% test accuracy in 493.6s

Decoupled Knowledge Distillation (DKD)

- **Setup**

- Same teacher–student setup as KD
- Distillation is applied on teacher and student outputs

- **Key Idea**

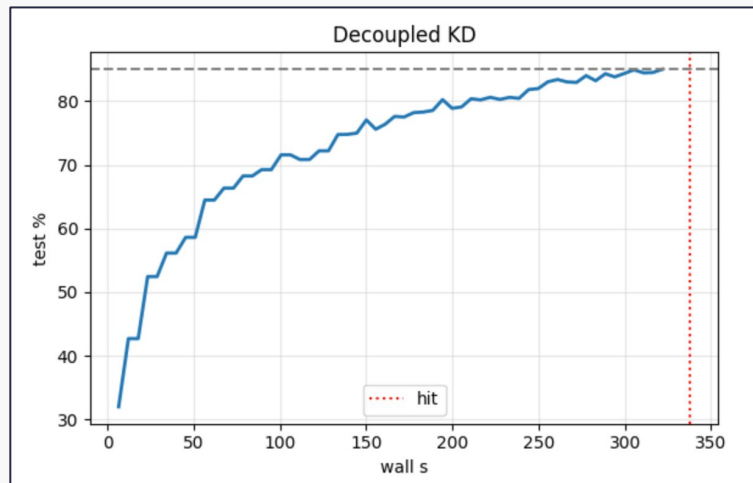
- Distillation is split:
 - TCKD (target class alignment)
 - NCKD (non-target “dark knowledge”)
- Controlled by hyperparameters:
 - α (target emphasis), β (non-target emphasis), and temperature (T)

- **Benefits**

- Separates correctness vs class relationships
 - Better control over learning dynamics
 - More stable and effective training

- **Limitation**

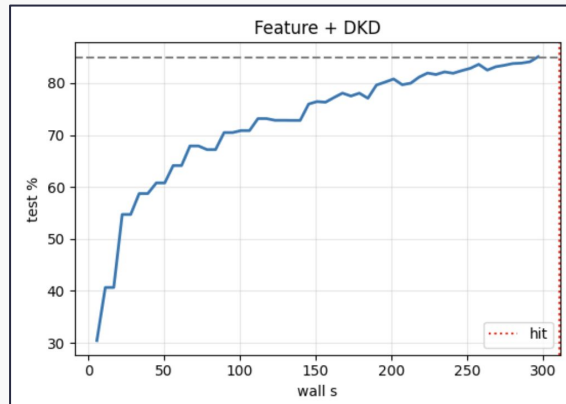
- Adds extra hyperparameters
- Requires tuning for best performance



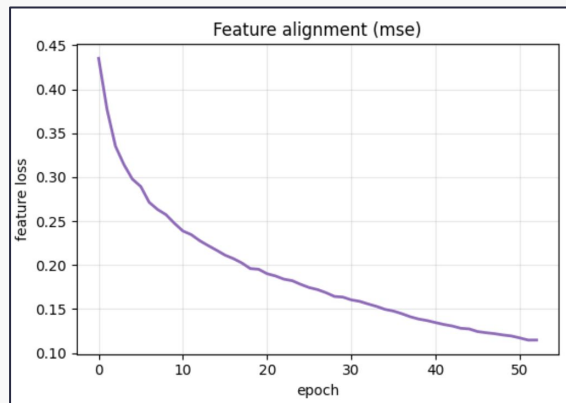
DKD reached 85.0% test accuracy in 337.81s

Feature Distillation + DKD

- **Setup**
 - Same teacher–student setup as KD
 - DKD + feature matching is combined
- **Key Idea**
 - Distills both:
 - Outputs (DKD)
 - Internal features (representations)
- **Benefits**
 - Features capture how inputs are represented
 - Provides deeper supervision signal
 - Faster convergence and better representations
- **Limitation**
 - More complex (extra loss + layer choices)
 - Higher computational cost



*Feature-Distillation:
reached 85.0% in
310.76s*



*Feature-Distillation
loss decreases to 0.12
over 50 epochs*

Conclusion

Takeaways

System optimizations only improve throughput and memory.

Distillation solves this problem by also improving learning efficiency and convergence behavior.

Implications

Provides a reproducible baseline for fast ViT training on CIFAR-10.

Next Steps

Create an open-source GitHub repository to find methods to further optimize knowledge distillation.

References

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<https://github.com/KellerJordan/cifar10-airbench>

<https://github.com/KellerJordan/modded-nanogpt/tree/master>